



Lifestyle interventions as adjuvant treatments for depression: An umbrella review of systematic reviews and meta-analyses

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ABSTRACT

Background: Depression is a high-prevalent mental health disorder that significantly affects quality of life and remains a leading global contributor to disability. Although pharmacological and psychological interventions are effective, many patients fail to achieve complete remission of depressive symptoms. Current evidence suggests that lifestyle-based interventions may offer a promising strategy as an adjuvant treatment for depression.

Objective: To evaluate the effectiveness of lifestyle-based interventions as adjuvant treatments for reducing depressive symptoms in adults.

Methods: We conducted an umbrella review of systematic reviews and meta-analyses of randomized controlled trials (RCTs) evaluating lifestyle-based interventions for adults with depression. Searches were performed in Medline/PubMed, PsycINFO, Web of Science, Cochrane Library, and Scopus from inception to March 2025. The review followed PRISMA 2020 guidelines and was prospectively registered in PROSPERO (CRD42024420246). Two reviewers independently screened studies, extracted data, and assessed methodological quality using AMSTAR-2, with discrepancies resolved through discussion with a third author. Findings were synthesised narratively.

Results: Forty-six systematic reviews were included. Exercise interventions showed the most consistent antidepressant effects, particularly supervised moderate-intensity aerobic and resistance training. Mind-body therapies demonstrated moderate-to-large benefits, especially in older adults. Dietary interventions showed small-to-moderate benefits and multicomponent interventions yielded small short-term effects, although their effectiveness was limited by heterogeneity and follow-up decline. The methodological quality was predominantly low in the included reviews.

Conclusions: Lifestyle-based adjuvant interventions show relevant antidepressant effects, particularly supervised moderate aerobic exercise and mind-body therapies. However, heterogeneity and low methodological quality limit conclusions, highlighting the need for robust trials to define optimal dosing, durability, and effectiveness.

1. Introduction

Depressive disorders are among the most prevalent mental health

conditions worldwide, affecting an estimated 3.8% of the global population (approximately 5% of adults), and constitute a leading cause of disability¹. Despite advances in pharmacological and psychological

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treatments, approximately 20–30% of individuals experience treatment-resistant forms of depression^{2,3}. This persistent challenge underscores the need to identify complementary and adjuvant strategies that can be integrated alongside standard treatments to enhance effectiveness, address residual symptoms, and target broader psychosocial, behavioural, and physiological dimensions of the disorder. These interventions aim to support first-line treatments by addressing modifiable risk factors, strengthening self-management capacities, and promoting long-term recovery and functional improvement. Such approaches are particularly relevant given the complex, multifactorial nature of depressive disorders, which involves interactions among biological vulnerability, psychological processes, lifestyle factors, and social determinants⁴.

Growing evidence highlights the significant role of lifestyle-related factors, such as low levels of physical activity (i.e., any bodily movement that increases energy expenditure), unhealthy dietary patterns, disrupted sleep patterns, and chronic stress⁵, in the onset, course, and persistence of depressive disorders⁶. These factors may influence physiological mechanisms including inflammation, oxidative stress, and neuroendocrine regulation, as well as emotional regulation and broader psychological functioning⁷.

Lifestyle medicine (LM) is an emerging and increasingly accepted discipline that focuses on the prevention, management, and, where possible, reversal of chronic conditions through evidence-based behavioural and lifestyle interventions targeting key modifiable determinants of health, such as physical activity, nutrition, sleep, stress management, and social and environmental influences^{8–10}. Within the conceptual framework of LM, lifestyle interventions can be defined as structured, non-pharmacological strategies aimed at modifying health-related behaviours and daily routines known to influence mental health, including exercise, dietary habits, sleep hygiene, and stress regulation¹¹.

Within this context, structured exercise, defined as planned, repetitive, and goal-oriented physical activity with a prescribed dose (e.g., frequency, duration, intensity), has been shown to support mood regulation, stress coping, and psychological well-being, while also contributing to broader health benefits¹². Mind-body exercise can be considered within this category and refers to practices that integrate physical movement with attentional, emotional, and cognitive component to promote both physical health and mental well-being. These approaches, including yoga, tai chi, and qigong, have been associated with favourable clinical outcomes in depression¹³. Additionally, healthier sleep patterns, characterised by adequate duration, regularity, and quality, are associated with a lower risk of depressive symptoms over time, highlighting the potential of sleep-focused strategies in the prevention and long-term management of depression¹⁴. Likewise, healthier dietary patterns, particularly those rich in whole and minimally processed foods such as the Mediterranean diet, have been associated with a lower risk of depressive symptoms, suggesting a potential preventive and therapeutic relevance^{15,16}. A recent meta-analysis assessed the preventive potential of lifestyle-based interventions in reducing depressive symptoms, finding a small but statistically significant effect ([Standardized Mean Difference (SMD)] = -0.29; 95% CI: -0.40 to -0.18). The study synthesized data from 27 randomized controlled trials, many of which included interventions focused on exercise, diet, sleep quality, and stress reduction. These findings underscore the clinical relevance of lifestyle interventions as a feasible strategy for the management of depression¹⁷.

The relationship between depression and lifestyle factors is bidirectional and self-reinforcing. Depressive symptoms such as anhedonia, fatigue, and psychomotor retardation often lead to reductions in both general physical activity and engagement in structured exercise, as well as poorer dietary choices and sleep disturbances, which in turn further exacerbate symptom severity and functional impairment^{18–21}. This reciprocal dynamic highlights the importance of integrated treatment approaches that address not only psychological and pharmacological targets but also behavioural and lifestyle-related determinants of

depression. Accordingly, interventions targeting exercise, diet, sleep, and stress regulation, delivered as single-component or multicomponent approaches, may play a complementary and adjuvant role by interrupting these maladaptive cycles and supporting long-term recovery¹⁸. From an integrative health perspective, lifestyle interventions represent a promising adjuvant approach to managing depression, as they target modifiable behavioural risk factors that contribute to its development and maintenance²². In addition to their preventive role, emerging evidence highlights the innovative potential of lifestyle-based interventions to significantly improve depressive symptoms even after the disorder has been diagnosed^{12,23}.

The present review examines the effectiveness of various lifestyle-based interventions, in the management of depression. Considering the multifactorial nature of depression and its strong association with modifiable lifestyle factors, an integrative approach that incorporates lifestyle modifications may be important; however, the specific characteristics and components of effective interventions remain insufficiently defined¹⁷. Despite this promising evidence, the existing literature shows substantial variability in intervention components, target populations, and methodological quality. Most interventions have been examined in isolation, with relatively few studies implementing or evaluating comprehensive multicomponent approaches. This heterogeneity highlights the need for a systematic synthesis of the evidence to identify which combinations of lifestyle interventions are most effective, and under what conditions.

Therefore, the aim of this study is to summarise the most current and robust evidence on the comparative effectiveness of lifestyle interventions for depression. To this end, we synthesise findings from systematic reviews and meta-analyses to support shared decision-making and to develop an evidence map that highlights current knowledge and guides future research directions.

2. Methods

We conducted an umbrella review of systematic reviews and meta-analysis of randomized controlled trials (RCT) to summarise the effectiveness of lifestyle-based interventions for depression, while accounting for the methodological quality of the included reviews. Umbrella reviews consolidate evidence from multiple systematic reviews and meta-analysis into a single synthesis, enabling comparison across intervention domains and supporting the identification of research gaps to inform clinical decision-making.

As referred to in [Supplementary Table S1](#), the findings are reported according to the Preferred Reporting Items for Systematic Review and Meta-Analyses (PRISMA 2020 checklist)^{24–25}, and adhered to the Measurement Tool to Assess Systematic Reviews (AMSTAR-2) guidelines²⁶ for assessing methodological quality. The protocol for this umbrella review was prospectively registered in the PROSPERO database (International Prospective Register of Systematic Reviews, under registration number CRD42024420246).

2.1. Search strategy

An initial search was conducted by two researchers (XG-G and AGR) between 1st February and 15th May 2023. A second search was performed in March 2025 to identify studies published from 15th May 2023 to March 2025. This update was conducted by three researchers (XG-G, PG-P, and AGR). The search strategy included the following terms: “depressive disorder”, “depressive disorder, major”, “depressive disorder, treatment resistant”, “motor activity”, “diet mediterranean”, and “lifestyle”, combined with the terms “systematic review” or “meta-analysis” in the title. Searches were conducted in the following databases: Medline/PubMed, PsycINFO, Web of Science, Cochrane Library, and Scopus. Language filters were applied to include studies published in English, Spanish, or Portuguese, with no date restrictions. Additionally, the reference lists of selected reviews were manually screened to

identify further eligible studies. The complete search strategy and the list of excluded references are provided in the [Supplementary Table S2](#).

2.2. Eligibility criteria

The PICOS framework (Population, Intervention, Comparison, Outcome and Study design) was used to define eligibility criteria:

2.2.1. Population

Adults aged 18 and older with a diagnosis of depression or depressive symptoms, assessed using a standardized diagnostic tool or validated scale. Reviews focusing on populations with clinical conditions that could significantly influence the results (e.g., pregnancy, cancer,

neurodegenerative disease, or psychotic disorders) were excluded.

2.2.2. Interventions

Reviews assessing one or more healthy lifestyle interventions (e.g., Mediterranean diet, structured exercise, mind-body exercise/therapies, sun exposure, sleep habits, or stress management), compared to a control group.

2.2.3. Comparators

Acceptable comparators included care as usual (CAU), no intervention, placebo, waitlist, attention controls, or other active interventions.

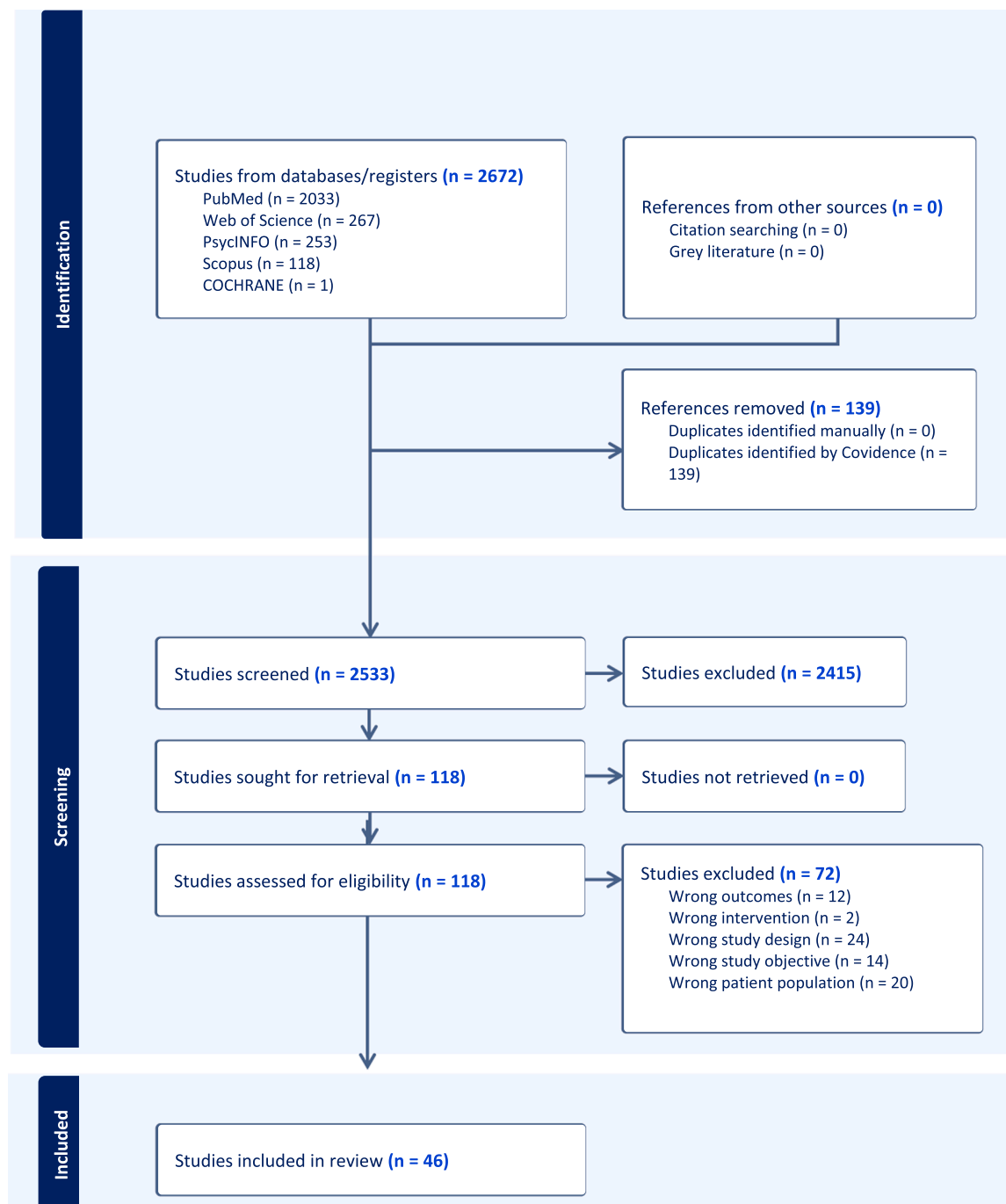


Fig. 1. Diagram Selection Flowchart (PRISMA). Source: Covidence.

2.2.4. Outcomes

Only reviews reporting depressive symptoms as the primary outcome were included.

2.2.5. Study design

We included only systematic reviews and meta-analysis of RCTs. Other designs were excluded.

2.3. Data extraction and management

Study selection was conducted in three phases: 1) removal of duplicates, 2) screening of titles and abstracts, and 3) full-text review of selected articles. No formal prioritisation strategy was applied to exclude overlapping reviews; all reviews meeting the eligibility criteria were included to provide a comprehensive overview of the field. Two reviewers (AGR and XG-G) independently screened and selected studies using the Covidence platform. Disagreements were resolved by consensus with a third reviewer (MB-V).

The following information was independently extracted from each included review: first author, year of publication, country, study design, number of included trials, type of lifestyle intervention, comparators, sample size and population characteristics, outcome measures, risk of bias assessment tool used, presence of GRADE assessment, and main findings with reported effect sizes (including confidence intervals and p-values, when it was described). A PRISMA flow diagram was developed to illustrate the study selection process (Fig. 1).

2.4. Risk of bias assessment

The methodological quality and risk of bias of the included systematic reviews were assessed using the AMSTAR-2 tool, which is specifically designed for systematic reviews of randomized and non-randomized healthcare interventions²⁶. The updated version (2017) includes 16 items focusing on critical domains such as protocol registration, inclusion criteria, comprehensive search strategies, duplicate study selection and data extraction, and justification for excluded studies²⁶. Unlike the original version, AMSTAR-2 does not generate an overall score but instead identifies critical methodological weaknesses that may affect the validity of the findings. Each included review was independently assessed by two reviewers, with disagreements resolved by consensus.

3. Results

3.1. Study selection

A total of 2672 records was retrieved through the database search. After removing duplicates, 2533 titles and abstracts were screened, resulting in 118 full-text articles assessed for eligibility. Ultimately, 46 systematic reviews evaluating lifestyle interventions for depression met the inclusion criteria and were included in this umbrella review (1.76% inclusion rate). Fig. 1 illustrates the flowchart of the study selection process across both databases. A list of the 72 excluded systematic reviews, along with reasons for exclusion, is provided in Supplementary Table S2.

3.2. Study characteristics

The characteristics of the included studies are summarized in Tables 1 and 2. Table 1 presents reviews involving participants with a clinical diagnosis of depressive disorder, while Table 2 presents reviews including participants with depressive symptoms at baseline.

The included evidence represents contributions from 18 different countries^{17,20,27–70}. The country with the highest number of publications was China, with 14 reviews, reflecting its strong research engagement in non-pharmacological interventions for depression^{17,29,}

^{37,41,47,55,57,58,61,62,64–67}. Several other countries also made substantial contributions: Australia, Spain, the United Kingdom, Germany, Brazil, the United States, Sweden, Denmark, Iran, and Canada. Additional single-country contributions came from the Netherlands, Norway, Greece, South Korea, Italy, Finland, and Colombia, highlighting the global interest in evaluating the effectiveness of lifestyle-based and mind-body interventions in the management of depressive symptoms, with particularly active participation from East Asia, Western Europe, and America^{20,27,28,30–36,38–40,42–46,48–54,56,59,60,63,68,69}.

Of the 46 systematic reviews included, 32 incorporated meta-analyses, several of which employed advanced analytical techniques such as subgroup analyses and, in some cases, meta-regression. Table 1 summarizes 38 reviews including participants with a clinical diagnosis of depressive disorder, while Table 2 presents 10 reviews that focused on individuals reporting depressive symptoms. Two reviews appear in both tables because they reported findings for both clinically diagnosed participants and individuals with depressive symptoms only. The methodological quality of the included studies was generally low, with most employing validated tools was the Cochrane Collaboration's Risk of Bias tool. Moreover, several reviews assessed the certainty of evidence using the GRADE framework, which further strengthened the reliability of their findings⁷¹.

The number of primary studies included in each review ranged significantly, from as few as five to as many as eighty randomized controlled trials. Reviews with the largest scopes, such as that by Tian et al., synthesized data from more than 70 RCTs⁶⁶.

The interventions examined were classified into four main categories: (1) exercise-based interventions, (2) mind-body therapies, (3) dietary interventions, and (4) multicomponent lifestyle interventions. Thirty reviews focused primarily on exercise interventions, of which eight also incorporated mind-body components such as yoga or qigong^{28,49,50,54,60,62,65,66}. Ten reviews exclusively evaluated mind-body therapies^{29,31,32,37,47,55,57,61,64,70}. Four reviews examined multicomponent lifestyle programs^{17,30,46,53}, and two focused specifically on dietary interventions^{59,63}.

3.3. Effects by type of intervention and population

3.3.1. Exercise-based interventions

Exercise-based interventions were the most frequently examined category across the thirty included reviews^{27–29,33,36,38–45,48–52,54,56,58,60,62,64,65,68}. These interventions involved a wide range of modalities, including aerobic training, resistance training, mixed programs, high-intensity interval training (HIIT), and neuromotor activities.

3.4. Depressive disorders (clinical population)

In populations with a clinical diagnosis of depressive disorder, aerobic exercise was the most consistently studied modality and demonstrated moderate-to-large reductions in depressive symptoms across age groups (SMD ranging from -0.31 to -1.04)^{52,62,65,72}. Resistance training interventions, such as strength or neuromuscular exercises, also showed moderate-to-large antidepressant effects, particularly among older adults^{49,58,72}. Several reviews examined mixed exercise programs combining aerobic and resistance components, as well as HIIT and neuromotor activities, which showed emerging efficacy; however, no consistent evidence suggested a clear superiority of combined modalities over single-modality approaches^{67,68}.

Exercise interventions were typically delivered three times per week, with individual studies ranging from 2 to 5 sessions per week^{42,56}. Session durations varied between 30 and 60 min, and most programs lasted between 4 and 16 weeks. Longer intervention durations (≥ 8 weeks) were associated with more robust outcomes⁶². Several reviews reported a nonlinear dose-response relationship, with optimal effects observed between 390 and 1000 MET-minutes per week^{62,65}. Moderate-intensity exercise delivered in supervised or group-based

Table 1
Summary of findings of included studies with diagnosed of depressive disorders' population.

Main author, Year of publication and Country	Design	Number of articles included in the reviews	Main intervention	Control groups	Size sample sociodemographic characteristics	Outcome measures	Risk of bias (YES/ NO) If is YES described which tool	GRADE (YES/ NO)	Main findings
Miller, 2020 Australia	Multivariate random-effects meta-analysis	16 (Included in the qualitative synthesis) /15 (Included in the quantitative synthesis)	EX (AE: walking, dancing; RT; MBE: Tai Chi, Qigong)	Wait-list, usual care or attention-control (people receive other interventions such as educational groups)	596 (321 intervention and 275 controls) in elderly people (>65 y) clinically depressed	BDI CES-D GDS GDS HRSD	YES (Cochrane Collaboration's Tool for Assessing Risk of Bias)	YES	AE SMD -0.31 (-0.45 , -0.17), RT -0.24 (-0.39 , -0.10), MBE (Tai Chi/Qigong) -0.36 (-0.52 , -0.20); MBE $\uparrow$ duration $\rightarrow$ $\uparrow$ effect; $p < 0.01$ – 0.001
Tsang, 2008 China	Systematic review	13	Mindful (MBE: Tai Chi, Qigong, Yoga) / Non-mindful EX (AE: pram walking, jogging, supervised EX)	Drug medication or placebo or partial yoga programme.	N = 631 adults clinically depressed	HRSD GDS BDI-II EPDS	YES	NO	11/12 studies +effect; non-mindful EX NS; long-term +effect in 3; meta-analysis not feasible due to limited high-quality studies
Cramer, 2013 Germany	Systematic review and meta-analysis	12/9	Complex yoga (MBE + breath control, meditation, and/or lifestyle advice) Exercise-based yoga (MBE; without breath control, meditation, or lifestyle advice)	CAU or active control	N = 619/452 Adults with depression (>18 y)	BDI HRSD GDS CDRS CES-D SDS EPDS	YES (Cochrane Back Review Group)	NO	Yoga vs CAU SMD -0.69 (-0.99 , -0.39); vs Relax -0.62 (-1.03 , -0.22); vs AE -0.59 (-0.99 , -0.18); I^2 0–86%; $p < 0.001$ – 0.004 ; effect medium
Lee, 2021 Canada	Systematic review and meta-analysis	23/22	Intervention: EX therapy (AE, RT, or both); MBE excluded.	CAU Active control group for example: health education sessions, stretching/toning light intensity exercise) or non-active control group.	N = 1025 Adults (<18 y) clinically depressed	HRSD BDI	YES (Cochrane Collaboration's Tool for Assessing Risk of Bias)	NO	Ex+CAU vs CAU SMD -0.62 (I^2 70%), severe pts -0.99 (I^2 0%); Sensitivity: low RoB -0.41 (I^2 52%), MDD -0.70 (I^2 72%); all $p < 0.001$
Schuch, 2016 Brazil	Meta-analysis	8	EX	Non-active control group, such as CAU, wait-list control conditions, placebo pills, or other social activities	N = 267 (138 participants in exercise arms and 129 controls) Older adults (>60 y) clinically depressed	HAM-D BDI GDS CSDD PHQ-9	YES (Cochrane Collaboration's Tool for Assessing Risk of Bias)	NO	Exercise large effect -0.90 (-1.51 , -0.29).
Catalán-Matamoros, 2016 Spain	Systematic review of reviews and meta-analyses	6/3	EX	CAU, wait list, no intervention or versus yoga	N = 1487 Elderly people (<60 y) clinically depressed	CES-D GDS	YES (AMSTAR)	NO	Exercise SMD 0.276 (0.070, 0.482); $p < 0.009$.
Danielsson, 2013 Sweden	Systematic review	14	AE first-line; pharmacological interventions excluded	Health education Pharmacotherapy CAU placebo exercise (relaxation, yoga) relaxation (Bobath balls, back massage, balance)	N = 1.139 Adults with depression (18–60 y)	BDI HRSD MADRS	YES (Cochrane Back Review Group criteria)	YES	AE vs antidepressants -0.06 (-0.36 , 0.23); AE vs PA 0.01 (-0.23 , 0.24); AE+CAU vs CAU -0.44 (-0.79 , -0.09)
Krogh, 2017 Denmark	Systematic review	35	EX vs control (CAU, WL, WL/Ni)	No intervention or CAU or an attention control using light exercise	N = 2.630 Adults clinically depressed	HRSD BDI HSCL-D	YES (Cochrane Handbook for Systematic Reviews of Interventions)	YES	Exercise vs control -0.66 (-0.86 , -0.46); $p < 0.001$
Zou, 2018 China	Meta-analysis	18/16	MBE: Tai Chi, Qigong, Yoga	Active and passive conditions	N = 844 Adults clinically depressed	HADS BDI CES-D	NO	NO* (PEDro)	Meditative movement -0.56 (-0.76 , -0.37); I^2 35.8%; $p < 0.001$

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Table 1 (continued)

Main author, Year of publication and Country	Design	Number of articles included in the reviews	Main intervention	Control groups	Size sample sociodemographic characteristics	Outcome measures	Risk of bias (YES/ NO) If is YES described which tool	GRADE (YES/ NO)	Main findings
Heissel, 2023 Germany	Systematic review and meta-analysis with meta-regression	41	EX; pharmacological interventions excluded	Non-exercising control group, such as CAU, wait-list control conditions or placebo pills	N = 2544 Adults clinically depressed	STAI QIDS BDI HRSD	YES (Cochrane risk of bias tool RoB2)	NO	Exercise interventions large $-0.946 (-1.18, -0.71)$; low RoB $-0.666 (-0.99, -0.34)$
Cramer, 2017 Germany	Systematic review	8	MBE – Exercise-based yoga	CAU or any active control intervention	N = 240 Adults with depression (>18 y)	BDI HRSD	YES	NO	Exercise vs AE NS; Yoga vs ECT ↑severity short-term
Schuch, 2016 Brazil UK	Meta-analysis	25	Supervised/unsupervised EX (AE, RT)	CAU, wait-list control conditions, placebo pills or other social activities.	N = 1.487 Adults with depression (>18 y)	MADRS BDI HAM-D	YES (Cochrane Collaboration's Tool for Assessing Risk of Bias)	NO	Exercise SMD 0.98 (0.68, 1.28); AE 1.04 (0.65, 1.43); moderate 1.33 (0.46, 2.19); vigorous 1.34 (0.43, 2.24); mixed 3.01 ($-0.61, 1.97$); supervised 1.26 (0.54, 1.97); 45 min $\times$ 3/wk $\times$ 4wks most effective Exercise g $-0.68 (-0.92, -0.44)$; follow-up g $-0.22 (-0.53, 0.09)$
Kvam, 2016 Norway	Meta-analysis	23	EX (AE, nonaerobic EX)	Placebo treatment, CAU, another kind of treatment (e.g., antidepressant medication, psychotherapy, or alternative interventions), no treatment, or be on a waiting list	N = 977 Adults (>18 y) clinically depressed	HRSD-17 BDI BDI-II SCL-90 HRSD-21 CESD	NO (Only publication bias assessment)	NO	Exercise g $-0.68 (-0.92, -0.44)$; follow-up g $-0.22 (-0.53, 0.09)$
Farah, 2016 USA	Systematic review	39/42	CBT, naturopathic therapy, biological interventions, EX	other treatment or CAU, waiting list or placebo	N = ~20000 Adults (>18 y) clinically depressed	Standardised depression scales	YES (Cochrane Collaboration's Tool for Assessing Risk of Bias)	YES	CBT/naturopathic/biological/exercise ↓depression; clinical response/remission unclear
Chen, 2019 China	Systematic review and network meta-analysis	91/32	Acupuncture, mirtazapine, herbal medicine, venlafaxine, EX, CBT, bupropion, fluoxetine, vortioxetine	Any inactive intervention (e.g., placebo, waiting list, sham treatment, no care), and above active treatments covered by the study	N = 10.991 adults clinically depressed	HRSD MADRS	YES (Cochrane risk-of-bias tool)	NO	Exercise OR 2.45 (1.05, 5.75) superior efficacy
Morres, 2019 Greece	Systematic review and meta-analysis	11	Intervention: AE. Duration (≤ 4 weeks), frequency ($< 3 / 3 / > 3$ d/week), intensity (low, moderate, vigorous), setting (indoor/outdoor), social format (group/individual), delivery (supervised/unsupervised)	comparisons: non-exercise comparators	N = 455 Adults (18–65 y) clinically depressed	HRSD BDI	NO	NO (PEDro)	AE 45 min $\times$ 3/week $\times$ 9.2wks g $-0.79 (-1.01, -0.57)$; p < 0.001
Nyström, 2015 Sweden	Systematic review	12	AE, anaerobic EX, or combination; detailed description of activity, intensity, duration, frequency	Controls: - Some type of EX, with stretching as the most used form. - CAU, i.e. pharmacotherapy or Psychotherapy -Health education.	N = ~1000 Adults clinically depressed	-	YES (Cochrane risk of bias tool)	NO	10/12 studies +effect; remission 26–86%; follow-up improvement rare

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Table 1 (continued)

Main author, Year of publication and Country	Design	Number of articles included in the reviews	Main intervention	Control groups	Size sample sociodemographic characteristics	Outcome measures	Risk of bias (YES/ NO) If is YES described which tool	GRADE (YES/ NO)	Main findings
Krogh, 2011 Denmark	Systematic review	13	EX (running, swimming, weightlifting)	Exercise vs a control group receiving no intervention or CAU or an attention control using light exercise	N = 2638 Adults clinically depressed	HRSD BDI PHQ-9 MADRS HSCL-D	YES (Cochrane Handbook for Systematic Reviews of Interventions)	YES	EX vs control -0.40 (-0.66 , -0.14)
Bridle, 2012 UK	Systematic review and meta-analysis	7	EX	CAU Waiting list Non-active control Attentional control	N = 667 Adults clinically depressed	GDS-15 BDI CES-D HRSD HSCL-20 CSDD	YES (Cochrane risk of bias criteria)	NO	EX Older adults SMD -0.34 (-0.52 , -0.17)
Fabricatore, 2011 USA	Systematic review and meta-analysis	31	Lifestyle modification, non-dieting, dietary counselling, diet-alone, EX program	-Waiting list standard - Advice/ printed materials, an -Attention-control intervention -No treatment.	N = 7937 Adults clinically depressed	BDI HRSD CES-D HSCL-90 Others subscales*	NO	NO	Lifestyle modification ↓depression $P < 0.001$; non-dieting $P < 0.002$; diet-alone $P < 0.002$; exercise-alone $P < 0.006$; dietary counselling NS.
Guo, 2019 China Korea	Meta-analysis	7	MBE: Qigong	control group as a comparison, such as a waitlist, CAU, pharmacological therapy, or walking.	N = 382 Adults clinically depressed	BDI HRSD CES-D	YES (Not Described)	NO	Qigong vs control Hedges' g -0.64 (-0.92 , -0.35); active -0.47 (-0.81 , -0.12); passive -0.80 (-1.23 , -0.37).
Mura, 2013 Italy	Systematic review	44	EX	Standard treatments (including antidepressant drugs or health education), CAU or placebo-control Attention group or wait list.	N = 1318 Adults clinically depressed	HRSD BDI GDS CES-D	Yes (Not described)	NO	Supervised physical exercise effective vs CAU/education/ medication
Nebiker, 2018 Switzerland Germany	Systematic review and meta-analysis	27	Supervised EX (AE, RT, functional EX training, MBE: Yoga/Tai Chi)	Behavioural therapy, light therapy, medication therapy, a placebo-group, or an inactive control group.	N = 1452 Adults clinically depressed	BDI HRSD	YES (Cochrane risk of bias tool)	NO (PEDro)	EX g -0.93 (-1.17 , -0.70); high-quality studies -0.83 (-1.13 , -0.54)
Seshadri, 2020 USA	Systematic review and meta-analysis	25	EX, MBE: Yoga/Tai Chi	Active control group or CAU.	N = 1.112 participants in the intervention groups and 971 in the control groups. Adults clinically depressed	BDI HRSD MADRS EPDS	YES (Cochrane risk of bias tool)	NO	EX/Yoga/Tai Chi g 0.63 (0.50 , 0.76); low RoB g 0.41 (0.14 , 0.66); vs active 0.45 (0.26 , 0.63); vs CAU 0.81 (0.59 , 1.30)
Sjösten, 2006 Finland	Systematic review	13	EX	Active controls, combined, attention and wait list control groups	N = 1264 Adults clinically depressed.	CES-D BDI HAM-D HRSD SDS	YES (Not described)	NO	EX effective vs control in older adults
Stanton, 2014 Australia	Systematic review	5	AE, RT, or combination; any duration.	Any other intervention including pharmacotherapy, education, psychotherapy,	- Adults clinically depressed.	Validation rating scales (BDI)	NO	NO (PEDro)	AE/RT 45–60 min $\times$ 3/week $\times$ 4–12wks effective; supervision recommended

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Table 1 (continued)

Main author, Year of publication and Country	Design	Number of articles included in the reviews	Main intervention	Control groups	Size sample sociodemographic characteristics	Outcome measures	Risk of bias (YES/ NO) If is YES described which tool	GRADE (YES/ NO)	Main findings
Cooney, 2013 UK	Systematic review	39	EX	other exercise program, or no intervention including wait list controls. Non-intervention or placebo Waiting list control TCC or TCC+exercise Other active controls (pharmacological, psychological and alternative therapies)	N = 2326 Adults (>18 y) clinically depressed	HDRS BDI	YES (Cochrane Collaboration's Tool for Assessing Risk of Bias)	YES	EX vs control −0.62 (−0.81, −0.42); vs psych −0.03 (−0.32, 0.26); vs alternative −6.40 (−10.2, −2.6); vs pharm −0.11 (−0.34, 0.12)
Noetel, 2024 Australia	Systematic review and network meta-analysis	218	EX (AE: walking, RT, MBE: Yoga)	Placebo, CAU or non-exercise	495 arms N = 14170 participants, general population, gender not specified	BDI, HAMD, PHQ-9	Yes (Cochrane RoB Tool)	YES	Active controls: walking/ jogging g −0.63 (−0.80, −0.46); yoga −0.55 (−0.73, −0.36); strength −0.49 (−0.69, −0.29); mixed AE −0.43 (−0.61, −0.25); Tai Chi/Qigong −0.42 (−0.65, −0.21); combined w/ SSRIs −0.55 (−0.86, −0.23); AE+psychotherapy −0.54 (−0.76, −0.32).
∞ Zhang, 2025 China	Systematic review & meta-analysis Until late 2023	18 RCTs	AE, RT, mixed EX	Non-exercise or CAU	N = 1830 participants / ~60% female Adults clinically depressed.	GDS, BDI, PHQ-9	YES (Cochrane ROB)	YES	AE middle-aged/older −0.41 (−0.60, −0.22); RT slightly more effective
Bizzozero-Peroni, 2025 Spain	Systematic review and meta-analysis	5	Mediterranean diet	Control group (no diet intervention)	N = 1507 participants, adults clinically depressed.	BDI, HRSD, PHQ-9 SF-36	YES (Cochrane Risk of Bias Tool)	YES	Mediterranean diet −0.53 (−0.90, −0.16)
Pérez Bedoya, 2023 Brazil Colombia	Systematic review and meta-analysis	9	EX (AE, MBE: Tai Chi/ Yoga); pharmacological interventions excluded	Second-generation antidepressants, cognitive behavioural therapy, body-mind exercise, or no exercise interventions	N = 678 adults clinically depressed aged 20–72 years; 211 men (31.12%) and 467 women (68.88%)	BDI, HRSD, CES-D	YES (Cochrane Risk of Bias Tool)	YES	Only EX(Tai Chi) −0.27 (−0.58, 0.04); P = 0.09
Zeng, 2023 China	Systematic review and meta-analysis	12	MBE: Tai Chi	Control group (no Tai Chi)	N = 731 participants, older adults clinically depressed	HRSD, BDI	YES, Cochrane Risk of Bias Tool	YES	Tai Chi older adults −1.21 (−1.59, −0.83); I ² 87.6%; p < 0.001
Tang, 2024 China	Systematic review and network meta-analysis	47	EX (AE: walking, RT, MBE: Yoga, Qigong, Tai Chi)	Placebo, CAU or non-exercise	N = 2895 older adults (≥60 years) clinically depressed.; exact % female not specified	BDI, HRSD	YES, Cochrane Risk of Bias Tool	YES	Walking −0.87 (−1.24, −0.51) to Tai Chi −0.66 (−1.14, −0.19); 30–60 min × 3–5/week × ≥8weeks most effective.
Swainson, 2023 Canada	Systematic review	5	Whole diet; pharmacological interventions excluded	No dietary intervention	N = 361 participants, 127 male (35,2%) and 234 female (64,8%)	PHQ-9, BDI	YES, risk of bias assessment conducted	YES	Dietary interventions (Mediterranean) improved mood; more research needed
Tian, 2024 China	Systematic review and Bayesian model-based network meta-analysis	46	EX (AE: walking, RT, MBE: Yoga, Qigong, Tai Chi); pharmacological interventions excluded	Placebo, CAU or non-exercise	2895 older adults (≥60 years); exact percentage of female participants not specified	BDI, HRSD, MADRS MET min/week dosage	YES (Cochrane Risk of Bias Tool)	YES	AE −0.93 (−1.25, −0.62); MBE −0.81 (−1.19, −0.42); mixed −0.77 (−1.20, −0.34); RT −0.76 (−1.24, −0.28)

(continued on next page)

Table 1 (continued)

Main author, Year of publication and Country	Design	Number of articles included in the reviews	Main intervention	Control groups	Size sample sociodemographic characteristics	Outcome measures	Risk of bias (YES/NO) If is YES described which tool	GRADE (YES/NO)	Main findings
Rossi, 2024 Brazil	Systematic review with meta-analysis	7	RT	Placebo or CAU	N = 2439 participants, adults and older adults clinically depressed	BDI, HRSD	YES(Cochrane Risk of Bias Tool)	YES	RT SMD 0.51 (0.30, 0.72); MDD 0.82 (−1.34, −0.30); $p < 0.002$
Martínez-Calderon, 2023 Spain	Systematic review with meta-analysis and meta-regression	23	MBE: Yoga-based	No intervention, waitlist or active controls	N = 1420 participants, adults clinically depressed.	BDI, PHQ-9	YES (Cochrane Risk of Bias Tool)	YES	Yoga vs control g −0.35 (−0.73, 0.03); I^2 82% NS; sensitivity excl. 3 outliers g −0.48 (−0.76, −0.19); I^2 57%. Qigong vs control g −0.49 (−1.59, 0.61); I^2 39% NS. Tai Chi vs control g −0.60 (−1.67, 0.47); I^2 75% NS. Supervised Yoga 60 min × 2–3/week × ≥8weeks most effective TCM exercise SMD −0.63 (−0.85, −0.41), $p < 0.01$. Supervised 30–60 min × 2–5/week × 8–24 weeks most effective. Tai Chi most effective for older adults.
Dong, 2023 China	Systematic review and network meta-analysis	30	MBE: TCM (Tai Chi, Qigong)	CAU, health education, non-Traditional Chinese Medicine exercise	N = 2806 participants, older adults clinically depressed	BDI, HRSD	YES (Cochrane Risk of Bias Tool)	YES	

Note. AE, Aerobic Exercise; BDI, Beck Depression Inventory; CAU, Care As Usual; CDRS, Cornell Dysthymia Rating Scale (Self-Report); CDS, Cornell Scale for Depression in Dementia; CES-D, Center for Epidemiologic Studies Depression Scale; CI, Confidence Interval; CDS, Cardiac Depression Scale; DASS, Depression Anxiety Stress Scale; ECT, Electroconvulsive Therapy; EPDS, Edinburgh Postnatal Depression Scale; EX, Exercise; GDS, Geriatric Depression Scale; g, Hedges' g; HADS, Hospital Anxiety and Depression Scale; HIIT, High-Intensity Interval Training; HRSD, Hamilton Rating Scale for Depression; HSCL-90, Hopkins Symptom Checklist-90; HSCL-D, Hopkins Symptom Depression Checklist; MADRS, Montgomery–Åsberg Depression Rating Scale; MBE, Mind–Body Exercise; MDD, Major Depressive Disorder; NI, No Intervention; NS, Not Significant; PA, Physical Activity; PHQ, Patient Health Questionnaire; POMS, Profile of Mood States; QIDS, Quick Inventory of Depressive Symptomatology (Clinician-Rated); RoB, Risk of Bias; RT, Resistance Training; SDS, Zung Self-Rating Depression Scale; SMD, Standardized Mean Difference; SSRIs, Selective Serotonin Reuptake Inhibitors; STAI, Spielberger State–Trait Anxiety Inventory; TCM, Traditional Chinese Medicine; WL, Waitlist; WL/NI, Waitlist/No Intervention.

Table 2

Summary of findings of included studies with depressive symptoms' population measured with standardized scales.

Main author, Year of publication and Country	Design	Number of articles included in the reviews	Main intervention	Control groups	Size sample sociodemographic characteristics	Outcome measures	Risk of bias (YES/NO) If is YES described which tool	GRADE (YES/NO)	Main findings
Wong, 2021 China	Meta-analytic review	75/50	Multicomponent lifestyle interventions (EX, diet, sleep, stress management)	CAU, wait list, no intervention or attention control	N = 8479 adults with depressive symptoms at baseline.	HADS CES-D DASS GDS BDI-II PHQ-9 CDS EPDS PHQ-8 MADRS	YES (Cochrane Collaboration's Tool for Assessing Risk of Bias)	NO	Lifestyleintervention vs CAU: d= 0.20 (0.12, 0.28), p < 0.001; vs WL/Ni: d= 0.22 (0.14, 0.30), p < 0.01; short-term 1–3mo d= 0.25 (0.15, 0.35); face-to-face > online; NS vs active controls
van Dammen, 2018 The Netherlands	Systematic review and meta-analysis	5	Lifestyle intervention, behavioural therapy, diet, EX, weight loss	Wait list, usual advice of healthy lifestyle and an active control (with exercise).	N = 224 (Reproductive age woman with BMI>25) with depressive symptoms	CESD POMS STAI	YES (Cochrane Collaboration's Tool for Assessing Risk of Bias)	NO	Lifestyle interventions overall: SMD −2.12 (−2.93, −1.32), p < 0.001
Young, 2018 Australia	Systematic review		Alcohol reduction Sleep improvement / insomnia reduction; ↑PA; Substance abuse reduction / cessation; Smoking cessation	Health Watch Generic Information Depression CBT programme Psycho-education Counselling Education and social support	N = 1964 adults with depressive symptoms at baseline.	PHQ-9 BDI MADRS			3/7 studies: post-intervention improvement; improvement depends on targeting lifestyle behaviour
Sjösten, 2006 Finland	Systematic review	13	EX	Active controls, combined, attention and wait list control groups	N = 1264 adults with depressive symptoms at baseline	CES-D BDI HAM-D HRSD SDS	YES (Not described)	NO	EX: effective for high baseline depression vs controls.
Xu, 2024 China	Systematic review and meta-analysis	11	MBE: Tai Chi, Yoga, Pilates	Non-exercise	N = 1005 participants, perimenopausal/postmenopausal women with depressive symptoms at baseline	PHQ-9 HADS BDI	YES (Cochrane Risk of Bias Tool)	YES	MBE overall: SMD −0.80 (−1.17, −0.44), P < 0.0001, I ² 79%
Dong, 2024 China	Systematic review and network meta-analysis	42	MBE (Tai Chi)	Non-exercise	N = 2974 participants, older adults with depressive symptoms at baseline	Standardized scales	YES (Cochrane Risk of Bias Tool)	YES	Tai Chi SMD −0.63 (−0.85, −0.41), p < 0.01 – most effective.
Tao, 2024 China	Systematic review and meta-analysis	34	MBE: Tai Chi, Yoga, Pilates; pharmacological interventions excluded; HIIT	Non-active control group	N = 1607 adults with depressive symptoms at baseline	BDI, HRSD	YES, Cochrane Risk of Bias Tool	YES	MBE: small effect vs control SMD 0.40 (0.20, 0.60), I ² 73%; vs other exercise NS SMD 0.15 (0.01, 0.30), I ² 10%; medium effect vs non-active control SMD 0.53 (0.21, 0.84), I ² 80%
Rossi, 2024 Brazil	Systematic review with meta-analysis	7	RT	Placebo or CAU	N = 2439 participants, adults and older adults with depressive symptoms at baseline	BDI, HRSD	YES, Cochrane Risk of Bias Tool	YES	RT moderate effect SMD 0.51 (0.30, 0.72), p < 0.001

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Table 2 (continued)

Main author, Year of publication and Country	Design	Number of articles included in the reviews	Main intervention	Control groups	Size sample sociodemographic characteristics	Outcome measures	Risk of bias (YES/NO) If YES described which tool	GRADE (YES/NO)	Main findings
Tian, 2024 China	Systematic review and meta-analyses	40	MBE, Yoga, Qigong, Tai Chi	No intervention or active controls	5536 participants, older adults with depressive symptoms at baseline	BDI, HRSD, MADRS	YES, Cochrane Risk of Bias Tool	YES	Yoga: SMD 0.55 (0.35, 0.76) – most effective; Qigong SMD 0.52 (0.11, 0.92); Tai Chi SMD 0.42 (0.13, 0.71).
Tian, 2024 China	Systematic review and meta-analyses	80	EX (AE, RT, MBE)	Placebo, CAU or non-exercise	5536 participants, older adults with depressive symptoms at baseline, 68.5% female	BDI, HRSD, MADRS, MET	YES, Cochrane Risk of Bias Tool	YES	Older adults: RT SMD –0.68 (95% CI –0.90, –0.46); MBE SMD –0.54 (95% CI –0.72, –0.37); AE SMD –0.31 (95% CI –0.50, –0.13); mixed EX SMD –0.23 (95% CI –0.44, –0.01)

Note. AE, Aerobic Exercise; BDI, Beck Depression Inventory; CAU, Care As Usual; CDRS, Cornell Dysthymia Rating Scale (Self-Report); CDS, Center for Epidemiologic Studies Depression Scale; CI, Confidence Interval; CDS, Cardiac Depression Scale; DASS, Depression Anxiety Stress Scale; ECT, Electroconvulsive Therapy; EPDS, Edinburgh Postnatal Depression Scale; EX, Exercise; GDS, Geriatric Depression Scale; g, Hedges' g; HADS, Hospital Anxiety and Depression Scale; HIIT, High-Intensity Interval Training; HRSD, Hamilton Rating Scale for Depression; HSCL-90, Hopkins Symptom Checklist-90; HSCL-D, Hopkins Symptom Depression Checklist; MADRS, Montgomery-Åsberg Depression Rating Scale; MBE, Mind-Body Exercise; MDD, Major Depressive Disorder; NI, No Intervention; NS, Not Significant; PA, Physical Activity; PHQ, Patient Health Questionnaire; POMS, Profile of Mood States; QIDS, Quick Inventory of Depressive Symptomatology (Clinician-Rated); RoB, Risk of Bias; RT, Resistance Training; SDS, Zung Self-Rating Depression Scale; SMD, Standardized Mean Difference; SSRIs, Selective Serotonin Reuptake Inhibitors; STAI, Spielberger State-Trait Anxiety Inventory; TCM, Traditional Chinese Medicine; WL, Waitlist; WL/NI, Waitlist/No Intervention.

formats tended to produce the most consistent benefits ^{52,54}. Professionally supervised programs were also associated with greater adherence, larger symptom reductions, and higher completion rates compared with unsupervised interventions ^{34,42,49}.

3.5. Depressive symptoms (non-clinical populations)

In populations presenting with depressive symptoms, resistance training exercise emerged as the modality associated with the largest reductions in depressive symptoms, particularly among adults and older adults with higher baseline symptom severity. This suggests that structured, moderate-intensity exercise focused on strength and body awareness may be especially beneficial in this population ^{51,66,68}.

3.6. Mind-body interventions

Mind-body interventions constituted the second most frequently examined category of lifestyle-based interventions. These approaches include practices such as yoga, tai chi, and qigong, which integrate physical movement with attentional, emotional, and self-regulatory components. A total of ten systematic reviews exclusively evaluated mind-body interventions ^{29,31,32,37,47,55,57,61,64,70}, and an additional eight reviews incorporated mind-body components within broader exercise-based programs ^{28,49,50,54,60,62,65,66}.

3.7. Depressive disorders (clinical population)

In populations with a formal diagnosis of depressive disorder, mind-body therapies demonstrated moderate-to-large effects on depressive symptoms, with reported SMDs ranging from –0.45 to –1.21 ^{31,47,61}. Yoga was the most frequently studied modality and showed consistent efficacy across different age groups and clinical settings (SMD= –0.69) ^{31,32}. Several reviews highlighted that yoga-based interventions were particularly effective when delivered in structured group formats and with moderate session frequency. Tai chi was especially effective among older adults, particularly in interventions lasting longer than 24 weeks and employing standardized routines such as the 24-style format ⁶¹. These longer and more structured interventions appeared to yield more sustained benefits. Qigong, although less frequently investigated, also demonstrated moderate antidepressant effects, particularly when delivered in dynamic, movement-based formats rather than static or meditative-only forms (SMD= –0.64) ⁴⁷.

3.8. Depressive symptoms (non-clinical populations)

Across the included systematic reviews and meta-analyses, mind-body exercise interventions demonstrated moderate-to-large effects in reducing depressive symptoms in subclinical populations, particularly among older adults and perimenopausal/postmenopausal women. Pooled effect sizes ranged from moderate (SMD = –0.54 to –0.63) to large (SMD = –0.80) ^{55,57,64,65}. Among modalities, Tai Chi, yoga, and qigong were all associated with symptom improvements, with yoga and Tai Chi frequently ranking among the most beneficial. Network meta-analytic evidence further suggested that mind-body exercises were among the most effective exercise modalities, often outperforming aerobic exercise and showing comparable effects to resistance training. Overall, these findings indicate that mind-body therapies may yield clinically meaningful short-term improvements in depressive symptoms in subclinical populations ⁶⁴.

3.9. Multicomponent lifestyle interventions

Multicomponent lifestyle interventions combined two or more behavioural domains, typically including exercise, dietary counselling, sleep hygiene, stress management, mindfulness-based practices, and/or psychoeducation (face-to-face, telephone or online formats). Four

systematic reviews focused specifically on these approaches ^{17,30,46,53}.

3.10. Depressive disorders (clinical population)

Only one systematic review with meta-analysis examined multi-component lifestyle interventions in clinically depressed populations ⁴⁶. This review, which included 31 studies and 7937 adults, reported significant reductions in depressive symptoms for lifestyle modification, non-dieting, diet-only, and exercise-only interventions ($p < 0.01$), whereas dietary counselling alone was not effective ($p = 0.30$).

3.11. Depressive symptoms (non-clinical populations)

Three reviews examined multicomponent lifestyle interventions in non-clinical populations presenting with depressive symptoms at baseline. A large meta-analytic review found small but statistically significant short-term reductions in depressive symptoms compared to care as usual ($p > 0.001$; $d = 0.20$) and wait-list/no intervention ($p > 0.01$; $d = 0.22$) with effects maintained up to 3 months but not in the medium- or long-term follow-up. This review also showed that face-to-face delivery formats were more effective than online interventions ¹⁷. Another review reported a large pooled effect in women with depressive symptoms ($SMD = -2.12$), although this estimate was based on a small number of trials ($N = 224$) ³⁰. Finally, the third review found that only studies achieving actual changes in targeted lifestyle behaviours showed improvements in depressive symptoms, suggesting that behavioural change could be a key mechanism underlying intervention effect ⁵³.

3.12. Dietary interventions

Dietary interventions were the least frequently examined category, with only two systematic reviews exclusively focusing on nutrition-based approaches ^{59,63}. These reviews primarily investigated whole-food-based dietary patterns, particularly those aligned with the Mediterranean diet, which emphasize a high intake of fruits, vegetables, whole grains, legumes, nuts, and healthy fats.

3.13. Depressive disorders (clinical population)

In clinically diagnosed populations, dietary interventions were associated with small-to-moderate reductions in depressive symptoms ($SMD = -0.53$). Both reviews highlighted that greater adherence to healthier dietary patterns was linked to improvements in mood and overall psychological well-being, potentially mediated by anti-inflammatory and antioxidant mechanisms ^{59,63}.

3.14. Comparators

Regarding control conditions, CAU was the most frequently employed comparator ^{31,33,35,36,38,54,72}, followed by the wait-list control ^{20,28,44,45,73}. Several reviews also included active control conditions, such as health education, light stretching, or relaxation exercises, designed to account for nonspecific therapeutic effects while avoiding structured moderate-intensity exercise ^{33,43,49}. Placebo interventions, such as pills or treatments, were less commonly used ^{37,48}. Other noteworthy comparators included pharmacotherapy and structured psychotherapy, which were less frequent but clinically relevant. These comparisons often showed improvements of a similar magnitude in some comparisons ^{29,40,41,52}.

3.14.1. First-line versus adjuvant use of lifestyle-based interventions

Most of the included reviews conceptualized lifestyle-based interventions primarily as adjuvant treatments, delivered alongside pharmacological and/or psychological care. However, five reviews explicitly examined these interventions as first-line treatments, particularly in populations presenting with depressive symptoms rather than a

formal clinical diagnosis ^{35,38,60,63,70}.

3.15. Sample sizes and participant characteristics

Sample sizes varied substantially across the included reviews, ranging from 277 to over 20,000 participants. Reviews focusing on multicomponent and mind-body interventions tended to include larger cumulative samples. The most commonly used instruments to define the target populations and assess depressive symptoms were the Beck Depression Inventory (BDI), the Centre for Epidemiologic Studies Depression Scale (CES-D), the Hamilton Depression Rating Scale (HAM-D), the Geriatric Depression Scale (GDS), and the Patient Health Questionnaire-9 (PHQ-9). Across studies, women represented most participants, with proportions frequently exceeding 60% ⁵⁷.

3.16. Risk of bias assessment

The methodological quality of the included reviews, as assessed by the AMSTAR-2 tool ²⁶, showed considerable variability. Of the 46 reviews, 6 were rated as high quality, 4 as moderate quality, 7 as low quality, and 27 as critically low quality. While several high-quality reviews demonstrated rigorous methodology, particularly those employing comprehensive literature searches, duplicate study selection and data extraction, and appropriate risk of bias assessment ^{55,56,58}, a substantial proportion was rated as having low or critically low confidence due to methodological limitations. These limitations included the absence of prior protocol registration, insufficient reporting of excluded studies with justification, and lack of detailed consideration of funding sources for included trials. Additionally, although many reviews employed suitable statistical methods for meta-analyses, fewer adequately assessed publication bias or accounted for the impact of study-level risk of bias on pooled results. Overall, only a minority of reviews achieved a high confidence rating, while most were classified as low or critically low, underscoring the need for more robust methodological standards in future systematic reviews and meta-analyses on lifestyle interventions for depression. Tables 3 and 4 present a visual summary of the methodological quality and risk of bias across the reviews included.

4. Discussion

This umbrella reviews synthesised evidence from 46 systematic reviews and meta-analyses assessing lifestyle-based interventions for the adjuvant treatment of depression. This approach is consistent with current recommendations supporting integrated biopsychosocial treatments for depression, as combining biological, psychological, and social strategies can enhance treatment effectiveness in a shorter timeframe and contribute to improved long-term outcomes ⁷⁴. Physical exercise and mind-body therapies exhibit the most consistent and clinically meaningful benefits, while multicomponent interventions offer promising, though more modest, short-term improvements due to their integrative nature. Importantly, these interventions appear to be potentially safe, accessible, and particularly valuable as complementary treatments within multidisciplinary models of depression care. However, the substantial heterogeneity and predominantly low quality across studies limits the generalisability of the findings and precludes drawing firm conclusions regarding their overall effectiveness.

A key conceptual distinction emerging from this synthesis concerns the role of lifestyle interventions as adjuvant versus standalone treatments. Most of the evidence included in this umbrella review conceptualises these interventions as adjuvant strategies, designed to complement pharmacological and/or psychological treatments rather than replace them. In this context, lifestyle-based approaches may enhance overall treatment response by targeting residual symptoms, improving functional outcomes, and addressing behavioural and psychosocial mechanisms that are not fully captured by conventional

Table 3
Risk bias assessment.

	Miller, 2020	Wong, 2021	Tsang, 2008	Vin Dammen, 2018	Cramer, 2013	Lac, 2021	Schuch, 2016 PTE	Catalán-Matamoros, 2016	Danielsson, 2013	Krogh, 2017	Zou, 2018	Heisel, 2023	Cramer, 2017	Schuch, 2016	Kvam, 2016	Farah, 2016	Chen, 2019	Morres, 2019	Nysätröm, 2015	Krogh, 2011	Bridle, 2012	Fabricatore, 2011	Guo, 2019	Mura, 2013
1. Did the research questions and inclusion criteria for the review include the components of PICO?	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
2. Did the report of the review contain an explicit statement that the review methods were established prior to the conduct of the review and did the report justify any significant deviations from the protocol?	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
3. Did the review authors explain their selection of the study designs for inclusion in the review?	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
4. Did the review authors use a comprehensive literature search strategy?	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
5. Did the review authors perform study selection in duplicate?	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
6. Did the review authors perform data extraction in duplicate?	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
7. Did the review authors provide a list of excluded studies and justify the exclusions?	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
8. Did the review authors describe the included studies in adequate detail?	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
9. Did the review authors use a satisfactory technique for assessing the risk of bias (RoB) in individual studies that were included in the review?	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
10. Did the review authors report on the sources of funding for the studies included in the review?	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
11. If meta-analysis was performed, did the review authors use appropriate methods for statistical combination of results?	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
12. If meta-analysis was performed, did the review authors assess the potential impact of RoB in individual studies on the results of the meta-analysis or other evidence synthesis?	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
13. Did the review authors account for RoB in primary studies when interpreting/discussing the results of the review?	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
14. Did the review authors provide a satisfactory explanation for, and discussion of, any heterogeneity observed in the results of the review?	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
15. If they performed quantitative synthesis did the review authors carry out an adequate investigation of publication bias (small study bias) and discuss its likely impact on the results of the review?	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
16. Did the review authors report any potential sources of conflict of interest, including any funding they received for conducting the review?	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
AMSTAR-2: Overall confidence in the results of the review	C L	C L	C L	C L	C L	L	L	C L	C L	H	L	H	C L	C L	C L	C L	C L	C L	C L	C L	C L	C L	C L	L

Note. ● Yes; ● Partial Yes; ● No; H: High (No or one non-critical weakness); M: Moderate (More than one non-critical weakness); L: Low (One critical flaw with or without non-critical weakness); CL: Critically Low (More than one critical flaw with or without non-critical weakness).

Table 4
Risk bias assessment (continued).

	Nebiker, 2018	Seshadri, 2020	Sjosten, 2006	Stanton, 2014	Young, 2018	Cooney, 2013	Noel, 2024	Xu, 2024	Dang, 2024	Zhang, 2025	Bizzozero-Peroni, 2025	Pérez Bedoya, 2023	Zeng, 2023	Tang, 2024	Svalnson, 2023	Tian, 2024	Tao, 2024	Rossi, 2024	Martinez-Calderon, 2023	Tian, 2024	Tian, 2024	Dong, 2023
1. Did the research questions and inclusion criteria for the review include the components of PICO?	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
2. Did the report of the review contain an explicit statement that the review methods were established prior to the conduct of the review and did the report justify any significant deviations from the protocol?	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
3. Did the review authors explain their selection of the study designs for inclusion in the review?	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
4. Did the review authors use a comprehensive literature search strategy?	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
5. Did the review authors perform study selection in duplicate?	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
6. Did the review authors perform data extraction in duplicate?	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
7. Did the review authors provide a list of excluded studies and justify the exclusions?	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
8. Did the review authors describe the included studies in adequate detail?	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
9. Did the review authors use a satisfactory technique for assessing the risk of bias (RoB) in individual studies that were included in the review?	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
10. Did the review authors report on the sources of funding for the studies included in the review?	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
11. If meta-analysis was performed, did the review authors use appropriate methods for statistical combination of results?	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
12. If meta-analysis was performed, did the review authors assess the potential impact of RoB in individual studies on the results of the meta-analysis or other evidence synthesis?	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
13. Did the review authors account for RoB in primary studies when interpreting/discussing the results of the review?	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
14. Did the review authors provide a satisfactory explanation for, and discussion of, any heterogeneity observed in the results of the review?	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
15. If they performed quantitative synthesis did the review authors carry out an adequate investigation of publication bias (small study bias) and discuss its likely impact on the results of the review?	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
16. Did the review authors report any potential sources of conflict of interest, including any funding they received for conducting the review?	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
AMSTAR-2: Overall confidence in the results of the review	C L	L	C L	C L	C L	H	M	H	H	H	L	C L	C L	M	C L	C L	C L	C L	C L	M	M	L

Note. ● Yes; ● Partial Yes; ● No; H: High (No or one non-critical weakness); M: Moderate (More than one non-critical weakness); L: Low (One critical flaw with or without non-critical weakness); CL: Critically Low (More than one critical flaw with or without non-critical weakness).

therapies. However, a smaller subset of reviews evaluated lifestyle interventions, particularly exercise, as standalone or first-line strategies, mainly in populations with mild or subclinical depressive symptoms. In these cases, such interventions appear to function as low-intensity, or early-stage treatments, with outcomes comparable to those reported for psychotherapy and pharmacological interventions⁷⁵. This distinction has important implications for clinical decision-making, as it suggests that the appropriateness, expected benefits, and clinical role of lifestyle interventions depend on symptom severity, treatment context, and stage of care⁷⁵.

4.1. Efficacy of exercise as an adjuvant intervention for depression

Substantial body evidence supports the antidepressant effects of physical exercise across diverse populations with depressive disorders^{36,56,67,68,72}. Exercise modalities, predominantly aerobics but also including resistance and mixed modalities, consistently demonstrate moderate improvements in depressive symptom severity compared to usual care or non-exercise controls^{39,42,56}. Meta-analyses have estimated SMDs for exercise interventions in depressive symptoms ranging from -0.24 to -1.34 , which fall within the range of effect sizes reported for established treatments^{54,60}. Notably, the largest effect sizes were predominantly reported in reviews rated as low or critically low methodological quality, whereas reviews of higher methodological quality tended to report more modest effects, such as those observed in the high-quality review by Zhang (SMD = -0.41)⁵⁸. Notably, these estimates derive from both studies where exercise was used as a standalone intervention and those where it served as an adjunct to standard treatments. Evidence suggests that combined approaches, in which exercise is added to pharmacological or psychological therapy, tend to result greater improvements in depressive symptoms³³.

Aerobic exercise has emerged as the most extensively studied and effective modality, with multiple reviews highlighting its superiority in symptom reduction^{56,66}. Resistance training and neuromuscular exercises also show moderate benefits, particularly in older adults^{49,68}. Importantly, exercise appears effective across age groups: adults⁵⁶, and older adults^{45,72}, with structured exercise programs producing larger effects among institutionalized older adults⁵³.

4.2. Dose-response relationship and adherence

Emerging evidence points to a nonlinear dose-response relationship between exercise volume and depression outcomes, with optimal effects typically observed between 390 and 1000 MET-minutes per week^{62,65}. Exercise adherence appears to be relatively higher compared to other intensive interventions for depression, such as pharmacotherapy or structured psychotherapy, particularly those requiring frequent clinic visits or sustained cognitive engagement⁵². This enhanced adherence is often attributed to the flexibility and personalization of exercise programs, which are tailored to individual preferences regarding frequency, intensity, and setting. Moreover, group-based or supervised exercise interventions, especially those led by trained professionals, have been consistently associated with greater participant engagement, higher completion rates, and improved clinical outcomes, likely due to enhanced motivation, social support, and accountability mechanisms^{42,54}.

4.3. Applicability in clinical practice

Exercise exerts antidepressant effects through multiple biological and psychological pathways, including modulation of inflammatory markers, hypothalamic-pituitary-adrenal (HPA) axis regulation, neurogenesis, and improved neuroplasticity³⁸. Psychological benefits include enhanced self-efficacy, improved mood regulation, and stress resilience. In addition, exercise presents health benefits, such as cardiovascular and metabolic improvements⁵⁶.

Recent innovations, such as virtual reality-based exercise interventions, have shown promising results in improving depression and quality of life, particularly in older adults⁷⁶. Additionally, high-intensity interval training (HIIT) has demonstrated small-to-moderate benefits, expanding the range of effective exercise modalities for depression⁶⁷.

4.4. Efficacy of adjuvant mind-body therapies for depression

Mind-body therapies follow a holistic framework, emphasizing the bidirectional relationship between mental and physical health through self-regulation strategies, such as relaxation and meditation, that aim to enhance self-awareness and promote behavioural change. These interventions acknowledge the influence of emotional, cognitive, social, and spiritual factors on physiological processes. Mind body techniques frequently overlap with movement-based mindfulness practices like yoga, tai chi, and qigong, reflecting their shared emphasis on integrating attention, breath, and physical activity to improve overall well-being⁷⁷. There is a growing body of evidence supporting the use of mind-body therapies as effective adjuvant interventions for MDD^{28,29,31–33,37,47,49,50,53,55,57,60–62,64–66,69,70}. Recent systematic reviews and meta-analyses reinforce their clinical relevance by showing consistent reductions in depressive symptoms across diverse populations⁶⁴.

Multiple studies confirm that Yoga, Tai Chi, and Qigong are among the most effective mind-body therapies^{31,47,61,64}. For instance, yoga was repeatedly identified as a top-performing modality in recent network meta-analyses, with SMDs ranging from -0.36 to -0.69 in reducing depression severity^{37,64}, and comparable in effectiveness to aerobic exercise and psychotherapy; however, these included reviews rated as low and critically low methodological quality, except two of them that reports SMDs between -0.63 to -0.80 ^{55,57}. This aligns with earlier findings by Cramer et al. and Zou et al., who highlighted yoga's moderate-to-large short-term impact and its comparability to electroconvulsive therapy^{31,32,37}.

Tai Chi has also emerged as an effective therapy, particularly in older adults^{55,61}. Several studies reported effect sizes greater than -1.21 , with stronger outcomes in longer-duration interventions (≥ 24 weeks), especially those using the 24-style Tai Chi format (56, 62). In network analyses, Tai Chi was ranked highest among mind-body therapies for reducing both anxiety and depression symptoms in older adults⁶¹. However, these findings should be interpreted with caution, as the majority of the reviews reporting large effect sizes were rated as having low or critically low methodological quality.

Qigong has also shown significant antidepressant effects, although with slightly smaller effect sizes compared to yoga and Tai Chi. According to Tian et al., Qigong showed a SMD of -0.52 , with studies noting enhanced efficacy when combined with dynamic movement or aerobic training⁶⁴. These findings are consistent with those from Guo et al., who reported Hedges' $g = -0.64$ for Qigong against control groups; however, these effect estimates should be interpreted cautiously given that both reviews were rated as critically low methodological quality⁴⁷.

Furthermore, the inclusion of mindful components appears to enhance long-term emotional outcomes when compared to non-mindful physical activity like walking or running⁵⁵.

The two high-quality reviews assessing the effects of mind-body therapies in subclinical populations reported that these interventions may be promising for alleviating depressive symptoms in older adults and perimenopausal women. Regarding intervention preferences, the relative effectiveness of the practices followed the order Tai Chi, Yoga, and Qigong^{55,57}.

4.5. Applicability in clinical practice

Mind-body therapies enhance emotion regulation capacities, self-efficacy, and resilience, core factors in MDD treatment⁵⁷. Compared to pharmacological treatments or intensive psychotherapies, mind-body

therapies, generally, require fewer resources, particularly when delivered in group settings or via digital platforms which increases their cost-effectiveness. Emerging evidence supports the feasibility of internet-based and mobile mind-body interventions, which have also demonstrated effectiveness and feasibility in the clinical setting⁷⁸.

4.6. Efficacy of adjuvant multicomponent interventions for depression

Multicomponent lifestyle modification interventions demonstrated moderate short-term effects in reducing depressive symptoms, with SMDs ranging from -0.20 to -0.22 when compared to usual care or waitlist controls¹⁷. These improvements were maintained up to three months after the intervention¹⁷. This finding supports the hypothesis that the integration of two or three lifestyle components may positively modulate depressive symptomatology, particularly in individuals with depression^{17,30,46}.

One of these studies evaluated interventions promoting healthy diet, physical activity, and, in some cases, smoking cessation in general population samples. While the overall effect was small ($SMD = -0.184$), the effect sizes decreased when analyses were restricted to studies with low risk of bias, highlighting the critical role of methodological quality in interpreting outcomes⁵. Most of these reviews focused on post-treatment and short-term follow-up results²². In addition, substantial heterogeneity was observed in both the structure and duration of the interventions, which complicates direct comparisons across studies. This variability, arising from the differences in the lifestyle components included, limits the ability to precisely estimate their overall efficacy¹².

Evidence from some studies has shown that lifestyle interventions can significantly reduce both depressive symptoms and anxiety levels³⁰. These results support the potential of lifestyle-based strategies to deliver short-term mental health benefits¹⁷. This is further reinforced by findings from digital interventions, such as those reviewed by Young et al., which also demonstrate positive outcomes in the short term⁵³. Several authors have emphasized the need to tailor multimodal interventions to populations context to maximize their effectiveness^{5,30}.

4.7. Applicability in clinical practice

Summary, these findings suggest that lifestyle interventions, whether delivered face-to-face or online, can produce rapid improvements in depressive symptoms in the short term. However, their effectiveness appears to depend on study design, methodological quality, and the characteristics of the study population³⁰.

4.8. Other interventions

To a lesser extent, two reviews focusing exclusively on the effect of diet in the treatment of MDD were selected^{59,63}. These studies provide consistent evidence supporting the positive effects of dietary interventions, particularly Mediterranean diet, on depressive symptoms in individuals with MDD, although both reviews were rated as having low to critically low methodological quality according to assessment tool. They found that adherence to a Mediterranean diet significantly improved both self-reported and clinician-rated depression scores with a moderate antidepressant effect, especially when delivered by nutrition professionals⁵⁹. Despite some heterogeneity in study design and dietary protocols, both studies highlight the clinical relevance of diet quality as a therapeutic complement in depression management.

However, the interpretation of these findings must be contextualised within several important gaps identified across the included reviews. First, evidence regarding the durability and maintenance of benefits over the medium and long term remains limited, as most trials prioritised post-intervention or short-term outcomes⁷². Second, the systematic and detailed reporting of adverse events was frequently insufficient or absent, limiting the ability to draw robust conclusions regarding the

safety profile of lifestyle-based interventions⁷⁹. Third, intervention adherence was often inconsistently measured, and few studies explicitly examined how adherence varied according to the level of supervision, monitoring, or support provided^{80,81}.

4.9. Limitations

Multicomponent interventions, as well as single-component approaches, exhibit substantial heterogeneity in terms of structure, delivery methods, and the specific combination of lifestyle components (e. g., exercise, diet, smoking cessation, sleep hygiene). This variability complicates the accurate assessment and comparison of their efficacy across studies. Although many interventions have shown effectiveness in reducing depressive symptoms, particularly in more severe presentations, methodological improvements are needed to reduce bias, enhance comparability, and strengthen confidence in the findings. In particular, the inclusion of long-term follow-up assessments is essential to evaluate the sustainability of observed benefits over time.

An additional limitation of this review is the exclusion of grey literature, which may have led to the omission of relevant unpublished studies and increased the risk of publication bias. As an umbrella review, the synthesis of secondary evidence also limits access to primary-level data, and the methodological heterogeneity and potential overlap of primary studies across included reviews further warrant cautious interpretation of the findings. Moreover, restricting inclusion to reviews in which depressive symptoms were the primary outcome, while enhancing specificity and clinical relevance, may have excluded studies assessing depression as a secondary or co-occurring outcome, thereby limiting the scope of the evidence base.

Building on these limitations, future research should prioritise the development and evaluation of standardised lifestyle interventions using robust methodology and transparent reporting. Particular attention should be given to long-term effectiveness, adherence, safety and adverse event reporting, and individualisation, as well as to the role of multidisciplinary and digital delivery models. Beyond depressive symptoms, future studies should incorporate broader outcomes, including quality of life, cognitive and social functioning, and relevant biological markers, alongside comparative effectiveness and public health research to inform scalable and sustainable implementation.

5. Conclusions

Current evidence suggests that lifestyle-based interventions may be associated with clinically relevant reductions in depressive symptoms when implemented as adjuvant approaches alongside standard care. Supervised structured exercise and mind-body therapies appear particularly promising, while multicomponent lifestyle interventions tend to show smaller but potentially meaningful short-term effects. However, substantial heterogeneity in intervention components, intensity, and delivery formats, together with the predominantly low methodological quality of the included reviews, limits direct comparisons across studies and precludes firm conclusions regarding their relative effectiveness.

Overall, lifestyle-based interventions may represent complementary options within integrated models of depression care, but their routine implementation in clinical practice should be guided by cautious interpretation of the current evidence. Further high-quality, standardised research is needed to clarify long-term effectiveness, adherence, safety, and comparative effectiveness across intervention modalities.

Ethical approval

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CRediT authorship contribution statement

Garcia-Toro Mauro: Writing – review & editing, Validation, Methodology, Conceptualization. **Miquel Bennasar-Veny:** Writing – review & editing, Supervision, Methodology, Formal analysis, Conceptualization. **Aurora Garcia:** Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation, Formal analysis, Conceptualization. **Yañez Aina M:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Patricia García-Pazo:** Writing – review & editing, Writing – original draft, Validation. **Xandra Gonzalez-Garcia:** Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation, Formal analysis, Conceptualization.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used ChatGPT (OpenAI) to improve the readability and language of the manuscript. After using this tool, the authors reviewed and edited the content as needed and takes full responsibility for the content of the published article.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Author's contribution

XG-G; AGR; MB-V, AMY, MG-T: They prepare the conceptualization and introduction. XG-G, AGR, MB-V: They develop the method section, follow this process and independently screened, review and collect the articles. XG-G, AGR, PG-P: They write the original draft and create the tables and figures. MB-V, AMY, MG-T: They review, and editing based on the original draft with all sections. XG-G; AGR; MB-V, AMY, MG-T: They validation, review and editing the final version.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.ctim.2026.103336](https://doi.org/10.1016/j.ctim.2026.103336).

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